StealthComm - Encrypted Communication Device

Background

StealthComm is a discrete communication device allowing silent communication via morse code. Morse Code is a system used to represent letters, numbers, and punctuation through different combinations of dots and dashes. The system was invented in the 1830s by Samuel Morse for electric telegraph wires as these could only transmit electrical impulses (brief completion of a circuit) rather than full words or sentences.

While today's 'comms' provide a much faster and simpler method of conversation, morse code remains the most optimal way to communicate with utmost privacy and discretion.

The Logic

StealthComm relies on a ssh remote connection from the handheld device to an external computer. Once the connection is established, run morse_code_key.py to begin.

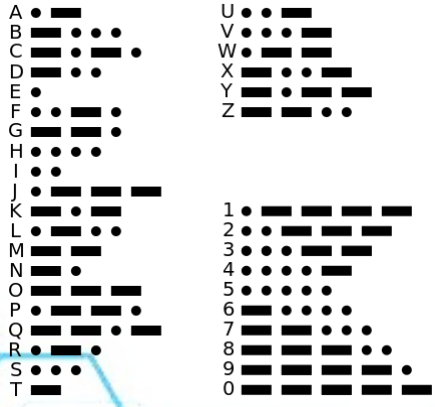
- The terminal should read "Connection secured. Send a message or type 'q' to leave: "
- Terminal will actively display any messages received from StealthComm. Note: Your message will be cancelled if you receive messages as you type. Press enter to confirm message. Note: You must wait for the prompt to re-appear to send another message.

StealthComm executes two functions running in parallel: Signal Out and Signal In.

SIGNAL OUT: Used to send a message in morse code from StealthComm device. Morse alphabet knowledge required.

- Using the button on your handheld device, input your desired message. Use the following guidelines to get started:
    - Press the button once to begin recording a message.
    - *Input Dot:* <= 0.4s press time
    - *Input Dash:* between 0.4-0.9s press time
    - *Separate letters:* between 1-1.5s gap time
    - *Separate words/input space:* between 1.5-3s gap time (aim for 2s)
    - Longer gap time = no action
    - Invalid letter (invalid combination of dots and dashes) = no action
    - To confirm and send the message, press for longer than 0.9 seconds.

Refer to Morse Alphabet below for translation:



https://www.codebug.org.uk/learn/step/540/morse-code-alphabet/

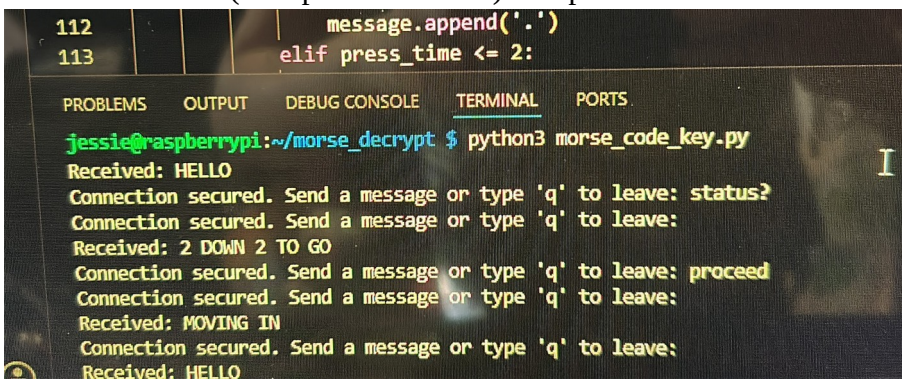
SIGNAL IN: Used to receive a message through StealthComm device sent by an external computer.

- No action required on StealthComm. Read and directly translate messages via the green LED indicator on your own device.
- Short flashes denote dots and long flashes denote dashes. Note that there will be 1 second pauses between letters and 1.7 second pauses between words.

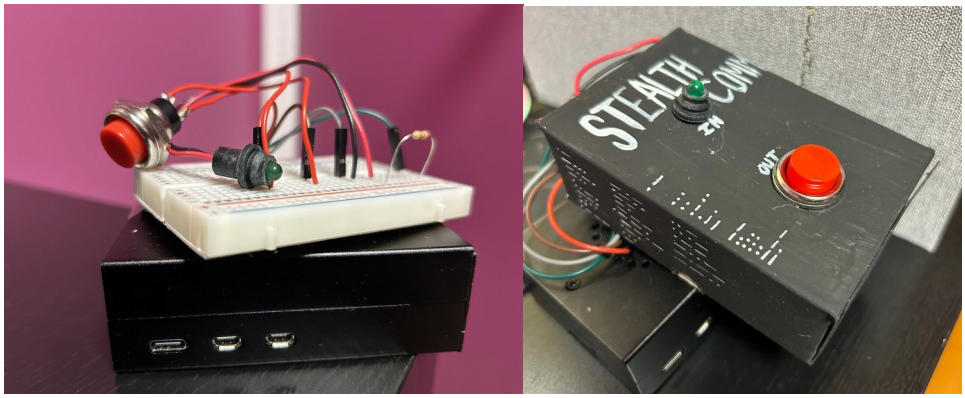
The Other End of the Line

- The person to whom you (StealthComm User) are communicating DOES NOT need to know morse code. The morse code encryption is made solely for the user who needs to send/receive information discretely from one end of the line.
- Once the message is sent from StealthComm (Person B), the code will be decrypted automatically and the recipient (Person A) will receive the resulting message in English via the terminal.
- Once the message is sent from the computer (Person A), the message will be encrypted automatically and the recipient (Person B) will receive the resulting message in morse code via the green LED indicator.

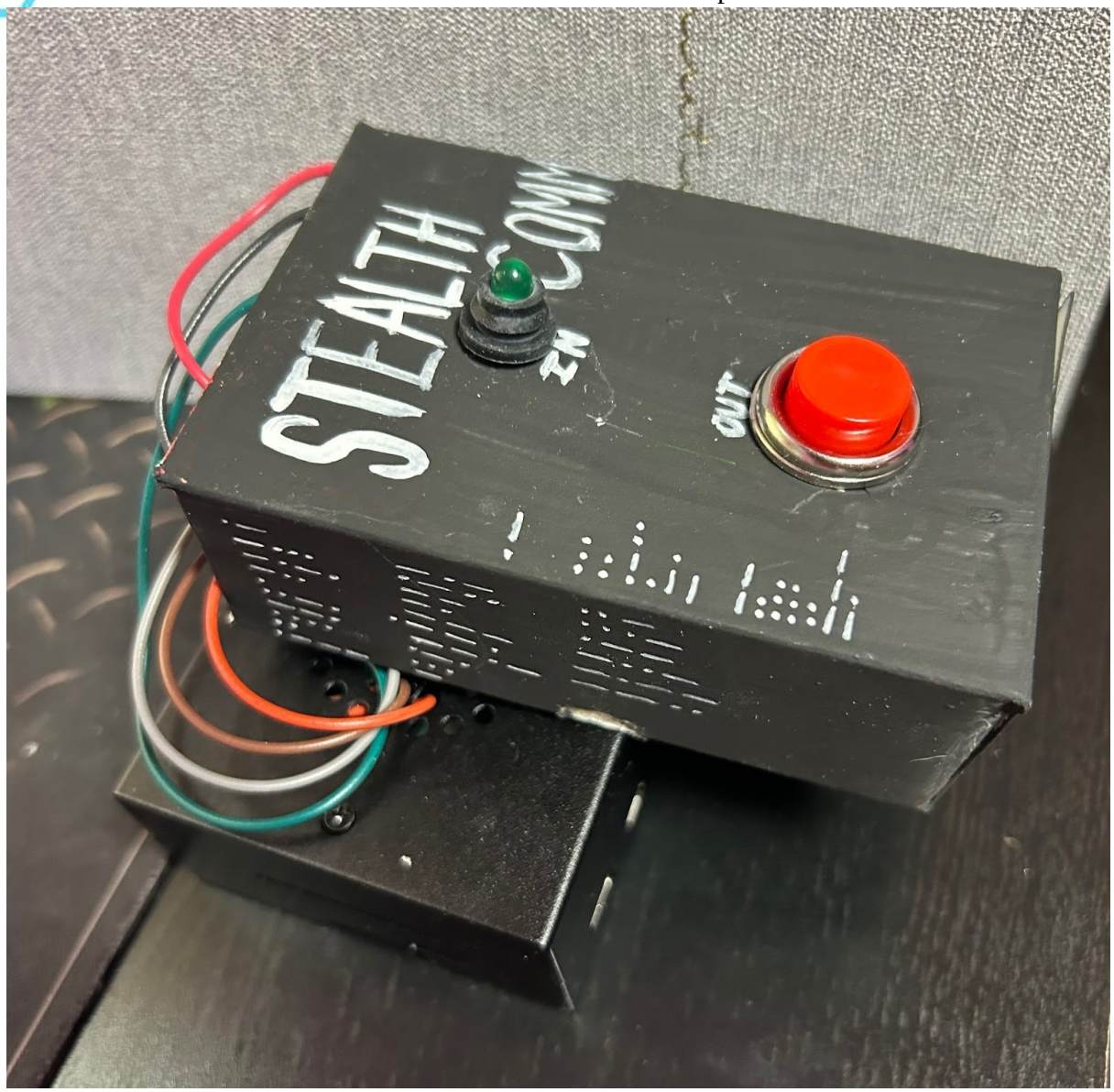
Person A Device (Computer Terminal) Sample

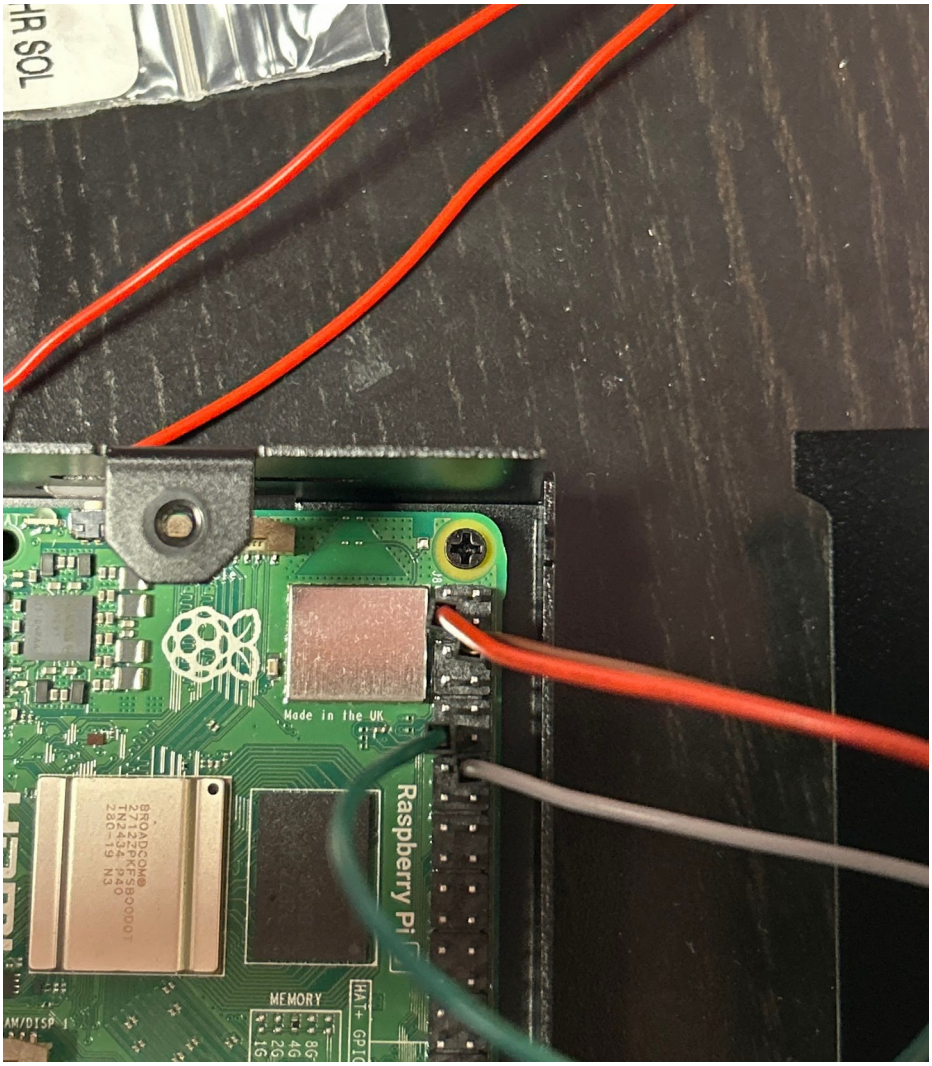


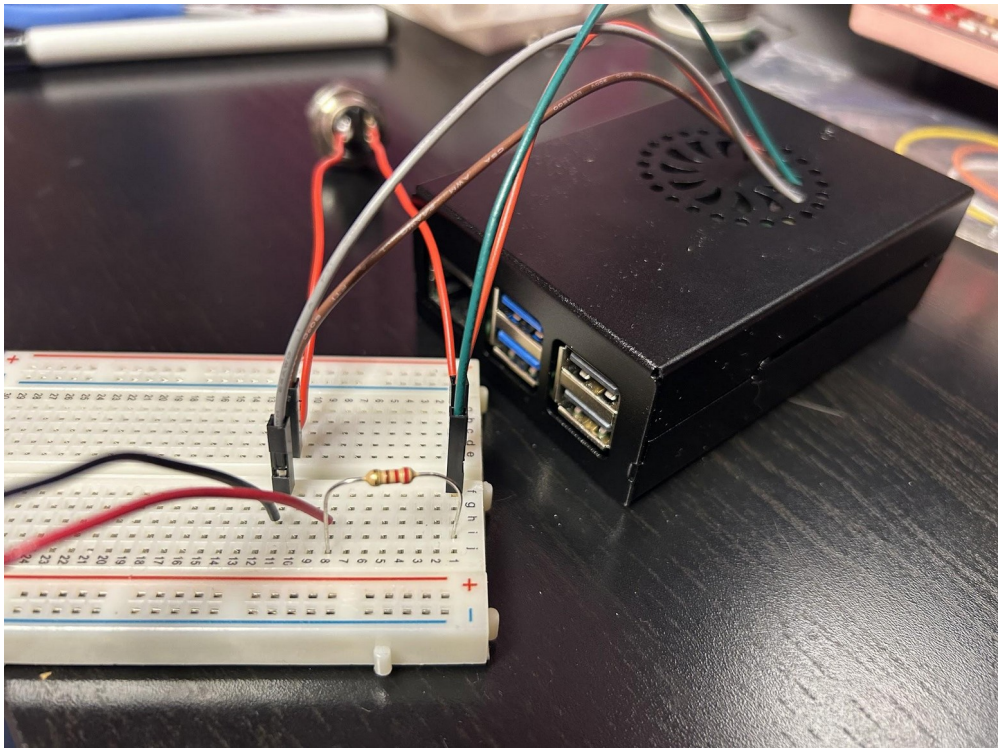
Person B Device (StealthComm) Prototype

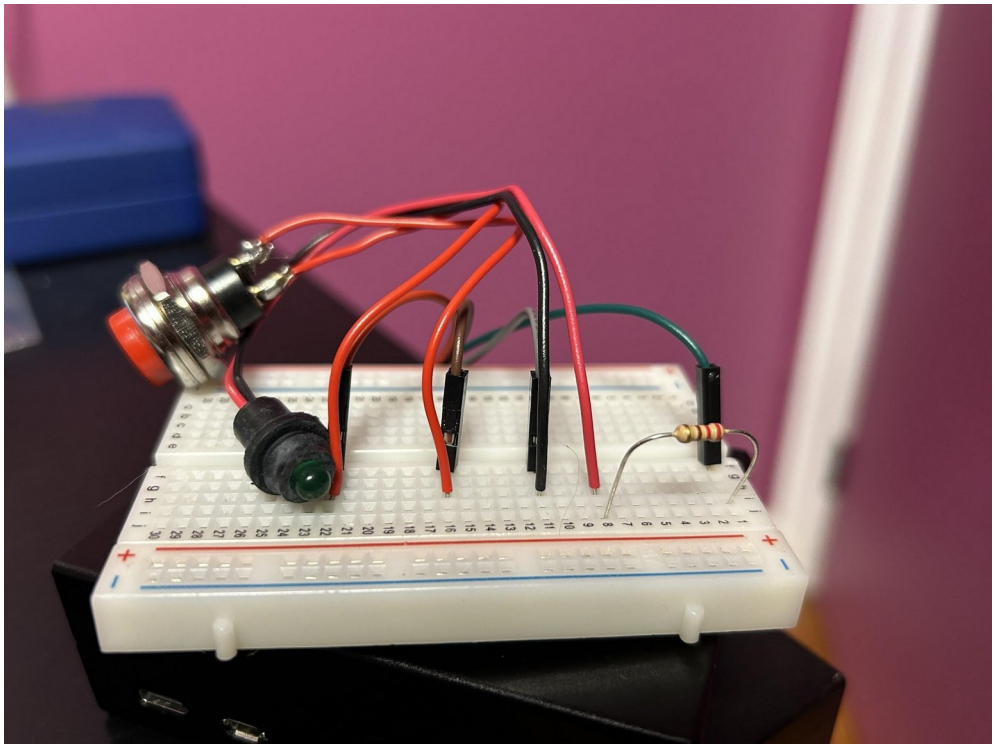


Additional Photos + Next Steps




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MEMORY





Next Steps

- Breadboard is to be replaced by male to female jumper wires to connect directly to Raspberry Pi.
- Pi to be eventually replaced by a smaller chip. Currently, the project shows a proof of concept but can easily be scaled down for convenience and added discreteness.